

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2023- 2024 (Odd Semester)

Active Learning Practices

Degree, Semester & Branch: B.E, V Semester & ECE A&B

Course Code & Title: CEC348 Remote Sensing

Name of the Faculty member: Dr.E.Emerson Nithiyaraj

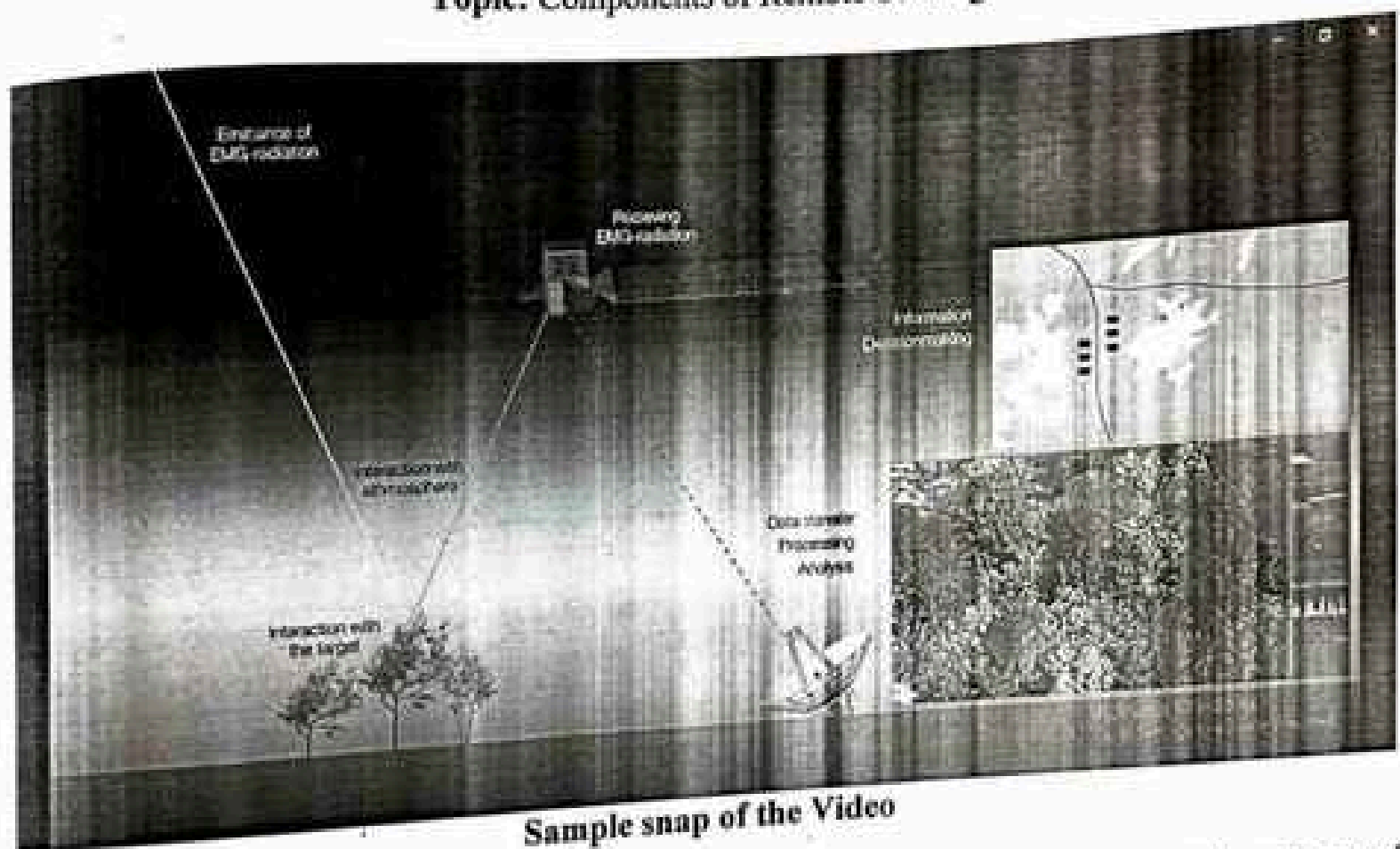
Date: 25.07.2023

Active Learning Practices Execution

UNIT I - REMOTE SENSING AND ELECTROMAGNETIC RADIATION

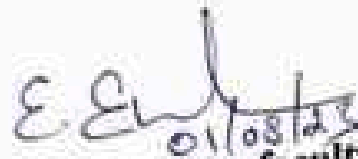
Activity: Animated Video

Topic: Components of Remote Sensing



Sample snap of the Video

The animated video on the "Components of Remote Sensing" delivered a dynamic and engaging learning experience by employing active learning strategies. Starting with an introduction to remote sensing's significance, the video utilized vibrant animations to elucidate key components, including various platforms and sensor types, the electromagnetic spectrum, resolution characteristics, image interpretation, geo-referencing, and data processing. This approach motivated students to think critically about the applications and implications of remote sensing technology in diverse fields.


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Signature of the faculty


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UNIT II - EMR INTERACTION WITH ATMOSPHERE & EARTH MATERIAL

Activity: One Minute Paper

Conducted Date: 10.08.2023

Topic: Energy Balance Equation

Handwritten notes on lined paper:

Energy balance 10/08/23

$E_s(\lambda) = E_a(\lambda) + E_r(\lambda) + E_t(\lambda)$

$E_s(\lambda) =$ Incident energy
 $E_a(\lambda) =$ Absorbed energy
 $E_r(\lambda) =$ Reflected energy
 $E_t(\lambda) =$ Transmitted energy

Conclusion:
 As short energy will only go through earth surface depends upon material type and also depends upon length, but short wave

Energy balance equation

The principle of conservation of energy can be used to describe energy transfer and energy interaction.

$E_s(\lambda) = E_a(\lambda) + E_r(\lambda) + E_t(\lambda)$

where:
 E_s - Incident energy
 E_a - Absorbed energy
 E_r - Reflected energy



Handwritten signature: E. Emerson

Signature of the faculty

Handwritten signature: 25/8/23

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Department of Electronics and Communication Engineering Academic Year 2023 – 2024 (Odd Semester)

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Name of the Faculty member: Dr.E.Emerson Nithiyaraj, AP/ECE

Innovative Practice Description

- Unit / Topic: Unit III / Classification of Satellites
- Course Outcome: CO3
- Topic Learning Outcome: TLO10
- Activity Chosen: Jigsaw
- Justification:

Jigsaw is a powerful active learning strategy that encourages students to become responsible for their own learning while also fostering teamwork and cooperation. It is particularly effective for complex topics and subjects that benefit from multiple perspectives and insights. It is often used in educational settings to promote collaborative learning and deeper understanding of complex subject matter.

- Time Allotted for the Activity: 50 minutes

- Details of the Implementation:

- Materials for the Activity:

The materials for the preparation of the students are shared one week before through LMS Canvas. Including the material, students are asked to refer to the following book and web content also.

Book Title/ Author/ Publisher/ Edition	Page No.
1. Thomas M. Lillesand, Ralph W. Kiefer and Jonathan W. Chipman, Remote Sensing and Image interpretation, John Wiley and Sons, Inc., New York, 2015.	283-379
Websites	
1. https://eos.com/blog/types-of-satellites/	
2. https://newspaceconomy.ca/2023/03/28/satellite-classification-taxonomy/	
3. https://www.isro.gov.in/Satellites.html	

- The topic on Classification of satellites was subdivided into five sub categories. The Students are divided into groups of six students.
- Each group is allotted with a name and it is known as Expert group EC1, EC2, EC3, EC4 & EC5
- Each expert group is allotted with one topic

- Communication Satellites
- Earth Observation Satellites
- Navigation Satellites
- Experimental Satellites
- Scientific Spacecraft
- The expert group members discuss the topic in detail for 20 minutes with the learning materials already posted in the course website – canvas. The Expert group members should go to other group and discuss the points (30 Minutes) to the other members.
- All the members in home group learnt about all the topics through expert group members. Finally one from each group should summarize the points of the topic Classification of satellites to the class (20 Minutes).
- Review of points and conclusion of the activity (5 minutes)
- Thus the concept of Classification of satellites is known to all by involving the students actively

- CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO6	PO7	PO12	PSO1
CO3	1	2	1	3	2	3	1	2
(1 – Low 2 – Moderate 3 – High)								

- PO / PSO mapped:

Innovative practice	PO 2	PO 4	PSO 1
	2	3	2
Justification for correlation	Students will be able to review and analyse complex problems at different domains of satellites. So Course outcome is mapped at level 2	In Jigsaw activity students will make effective discussions on the solutions that can be drawn for the complex problems, hence Course outcome is mapped at level 3	Students will be able to explore and adapt to the emerging trends in satellites. So Course outcome is mapped at level 1

- **Image / Screenshot of the practice:**



- **Reflective Critique:**

- ***Feedback of practice from students and other stakeholders:***

Students feel that it was interesting. They learn how to communicate with team members and work together. Students feedback was collected about the concept learned and activity used to learn. All the students replied positively about the activity and the learned concepts.

- ***Benefit of the practice:***

Every student got equal opportunity to come forward to take part in this activity. The success of the activity was evaluated by asking the same question in Internal Assessment test II. Outcome attainment would have increased due to innovative practice over conventional practice. Students easily understand the different domains of satellites and their respective applications.

- ***Challenges faced in implementation:***

Issues were faced while making the students to understand about Jigsaw rules.

References:

- Thomas M. Lillesand, Ralph W. Kiefer and Jonathan W. Chipman, Remote Sensing and Image interpretation, John Wiley and Sons, Inc., New York, 2015
- Dennis Roddy, "Satellite Communication", 4th Edition, Tata McGraw Hill International, 2006.
- <https://newspaceeconomy.ca/2023/03/28/satellite-classification-taxonomy/>

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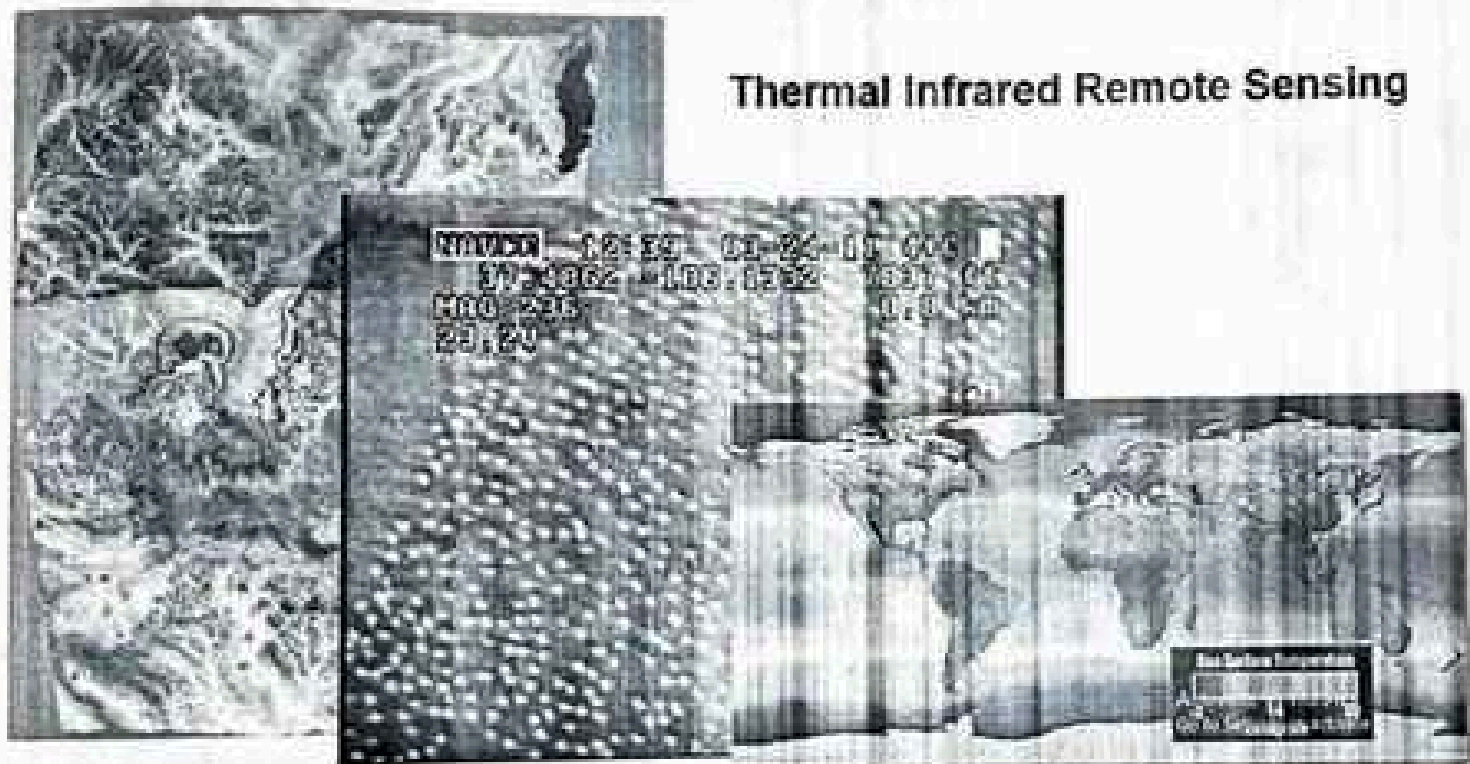
Date: 10.10.2023

Active Learning Practices Execution

UNIT IV - SENSING TECHNIQUES

Activity: Animated Video

Topic: Thermal Sensors



Sample snap of the Video

Commencing with a clear introduction to the role and significance of thermal sensors in remote sensing, the video utilized engaging animations to illustrate fundamental concepts, including the detection of thermal infrared radiation and the technology behind thermal sensors. By seamlessly combining visual appeal with thoughtful interactive features, this active learning approach not only elucidated the complexities of thermal sensors but also instilled a deeper appreciation for their indispensable role in remote sensing applications.


20/10/23
Signature of the faculty


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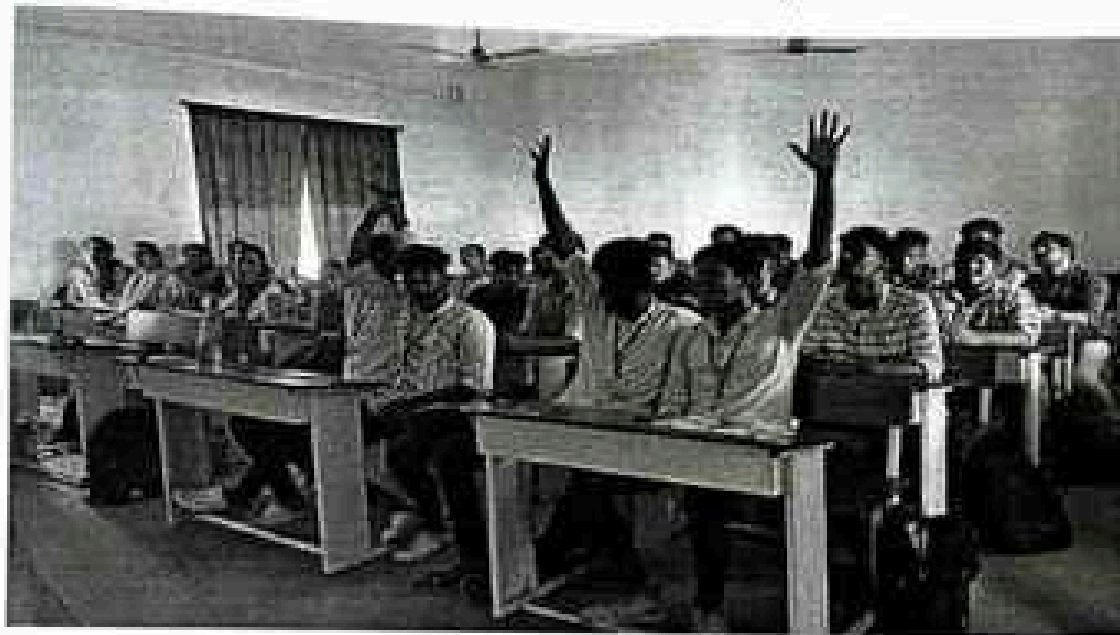
Date: 26.10.2023

Active Learning Practices Execution

UNIT V DATA PRODUCTS AND INTERPRETATION

Activity: Class Poll

Topic: Basic elements of Visual interpretation



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POLL QUESTIONS

1. What are the essential components of visual interpretation in remote sensing?
 - a) Colour, size, and shape
 - b) Visual cues, image features, and spectral information**
 - c) Satellite orbits, image acquisition, and cloud cover
 - d) GPS coordinates, elevation, and atmospheric conditions
2. Which visual cue is commonly used to distinguish different land cover types in remote sensing images?
 - a) Image resolution
 - b) Spectral bands
 - c) Shadows
 - d) Texture**
3. In visual interpretation, what role do image features play when analysing remote sensing data?
 - a) They provide information about the satellite's orbit.
 - b) They help identify spatial patterns and characteristics in the imagery.**
 - c) They determine the image resolution.
 - d) They are used for geographic coordinate transformation.
4. What is the significance of spectral information in visual interpretation?
 - a) It helps identify unique objects in the image.**
 - b) It describes the spatial arrangement of features.
 - c) It provides information about the satellite's altitude.
 - d) It relates to image brightness and contrast.
5. Which of the following is an example of a visual interpretation task in remote sensing?
 - a) Calculating the altitude of a satellite.
 - b) Identifying land cover types in an aerial photograph.**
 - c) Programming a GIS application.
 - d) Analysing weather patterns from satellite imagery.
6. When interpreting a satellite image, what can variations in colour or brightness indicate?
 - a) Differences in image acquisition time.
 - b) Cloud cover.
 - c) Land covers types or surface conditions.**
 - d) Satellite altitude.
7. What is the primary goal of visual interpretation in remote sensing?
 - a) To identify the satellite's technical specifications.
 - b) To assess the economic impact of land cover changes.
 - c) To extract meaningful information about the Earth's surface from imagery.**
 - d) To determine the distance between two points in an image.
8. When performing visual interpretation, what role does the human interpreter play in the process?
 - a) Supervising the satellite's data transmission.
 - b) Programming the image analysis software.
 - c) Applying knowledge and expertise to identify and classify features in the image.**
 - d) Controlling the satellite's sensors.